EED 101 INTRODUCTION TO ELECTRICAL ENGINEERING

SPRING 2017 **INSTRUCTOR: DR G NAVEEN BABU**

PROBLEM SET 1

REFERENCE TEXT BOOK: INTRODUCTORY CIRCUIT ANALYSIS, ROBERT BOYLESTAD, ***12TH EDITION***

CHAPTER 4:

PAGE NUMBERS: 131 AND 132

7,9,10,12,13,14,28,34,36,40,47,49

CHAPTER 5:

PAGE NUMBERS 176 TO 186

2B,2D,5,9,12,16,17,18B AND 18C, 20,21C,23B,24B,27B,27C,28C,28D,30B,40

7. $R = \frac{V}{I} = \frac{120\text{ V}}{2.2\text{ A}} = \mathbf{54.55\ \Omega}$

9. $R = \frac{V}{I} = \frac{120\text{ V}}{4.2\text{ A}} = \mathbf{28.57\ \Omega}$

12. $V = IR = (12\text{ A})(0.5\ \Omega) = \mathbf{6\ V}$

13. a. $R = \frac{V}{I} = \frac{120\text{ V}}{9.5\text{ A}} = \mathbf{12.63\ \Omega}$

b. $t = 2\cancel{\text{h}}\left[\frac{60\text{ min}}{1\cancel{\text{h}}}\right]\left[\frac{60\text{ s}}{1\text{ min}}\right] = 7200\text{ s}$

$$\begin{aligned} W &= Pt = VIt \\ &= (120\text{ V})(9.5\text{ A})(7200\text{ s}) \\ &= \mathbf{8.21 \times 10^6\ J} \end{aligned}$$

14. $V = IR = (2.4\ \mu\text{A})(3.3\text{ M}\Omega) = \mathbf{7.92\ V}$

28. $I = \dfrac{E}{R} = \dfrac{22\text{ V}}{16.8\text{ k}\Omega} = \mathbf{1.31\text{ mA}}$

$P = I^2R = (1.31\text{ mA})^2\ 16.8\text{ k}\Omega = \mathbf{28.83\text{ mW}}$

$W = Pt = (28.83\text{ mW})\left(1\text{ h}\left[\dfrac{60\text{ min}}{1\text{ h}}\right]\left[\dfrac{60\text{ s}}{1\text{ min}}\right]\right) = \mathbf{103.79\text{ J}}$

34. a. $P = EI$ and $I = \dfrac{P}{E} = \dfrac{0.4\times10^{-3}\text{ W}}{3\text{ V}} = \mathbf{0.13\text{ mA}}$

b. Ah rating = (0.13 mA)(500 h) = **66.5 mAh**

36. $P = EI = (220\text{ V})(30\text{ A}) = 6.6\text{ kW}$

$P_{HP} = \dfrac{6.6\text{ kW}}{746\text{ W/HP}} = \mathbf{8.85\text{ HP}}$

40. a. $W = Pt = (60\text{ W})(10\text{ h})\left(\dfrac{60\text{ min}}{1\text{ h}}\right)\left(\dfrac{60\text{ s}}{1\text{ min}}\right) = \mathbf{2.16\times10^6\text{ Ws}}$

b. 1 Ws = 1 J ∴ $\mathbf{2.16\times10^6\text{ J}}$

c. $W = Pt = (60\text{ W})(10\text{ h}) = \mathbf{600\text{ Wh}}$

d. $\dfrac{600\text{ Wh}}{1000\text{ W/1 kWh}} = \mathbf{0.6\text{ kWh}}$

e. Cost = (0.6 kWh)(11 ¢/kWh) = **6.6 ¢**

47. a. $P = EI = (120\text{ V})(100\text{ A}) = \mathbf{12\text{ kW}}$

b. $P_T = 5\text{ hp}\left[\dfrac{746\text{ W}}{\text{hp}}\right] + 3000\text{ W} + 2400\text{ W} + 1000\text{ W}$

$= 10{,}130 < 12{,}000\text{ W}$ **(Yes)**

c. $W = Pt = (10.13\text{ kW})(2\text{ h}) = \mathbf{20.26\text{ kWh}}$

49. $\text{kWh} = \dfrac{(200\text{ W})(4\text{ h}) + (6)(60\text{ W})(6\text{ h}) + (1200\text{ W})\ 20\text{ min}\ \dfrac{1\text{ h}}{60\text{ min}} + (175\text{ W})(3.5\text{ h}) + (250\text{ W})\ 2\frac{1}{6}\text{ h} + (30\text{ W})(8\text{ h})}{1000}$

$= \dfrac{800\text{ Wh} + 2160\text{ Wh} + 400\text{ Wh} + 612.5\text{ Wh} + 541.67\text{ Wh} + 240\text{ Wh}}{1000} = \mathbf{4.754\text{ kWh}}$

(4.754 kWh)(11¢/kWh) = **52.29¢**

Chapter 5

2b.

$R_T = 1.2\ \Omega + 2.7\ \Omega + 8.2\ \Omega = \mathbf{12.1\ \Omega}$

2d.

$R_T = 47\ \Omega + 820\ \Omega + 91\ \Omega + 1.2\ \text{k}\Omega = \mathbf{2158.0\ \Omega}$

5. a. $R_T = 10\ \Omega + 33\ \Omega + 56\ \Omega$, Reading = **99 Ω**
   b. $R_T = 2.2\ \text{k}\Omega + 0.82\ \text{k}\Omega + 1.2\ \text{k}\Omega + 3.3\ \text{k}\Omega$, Reading = **7.52 kΩ**

9. a. the most: R_3, the least: R_1
   b. R_3, $R_T = 1.2\ \text{k}\Omega + 6.8\ \text{k}\Omega + 82\ \text{k}\Omega = \mathbf{90\ k\Omega}$

   $$I_s = \frac{E}{R_T} = \frac{45\ \text{V}}{90\ \text{k}\Omega} = \mathbf{0.5\ mA}$$

   c. $V_1 = I_1R_1 = (0.5\ \text{mA})(1.2\ \text{k}\Omega) = \mathbf{0.6\ V}$, $V_2 = I_2R_2 = (0.5\ \text{mA})(6.8\ \text{k}\Omega) = \mathbf{3.4\ V}$, $V_3 = I_3R_3 = (0.5\ \text{mA})(82\ \text{k}\Omega) = \mathbf{41\ V}$, results agree with part (a)

d. $P_s = EI_s = (72\ \text{V})(1.8\ \text{A}) = \mathbf{129.6\ W}$
e. $P_{18\Omega} = V_3I_3 = (32.4\ \text{V})(1.8\ \text{A}) = \mathbf{58.32\ W}$

12. a. $I_m = \dfrac{E}{R_T} = \dfrac{36\ \text{V}}{4.4\ \text{k}\Omega} = \mathbf{8.18\ mA}$, $V_m = \dfrac{1}{2}E = \dfrac{1}{2}(36\ \text{V}) = \mathbf{18\ V}$

   b. $R_T = 1\ \text{k}\Omega + 2.4\ \text{k}\Omega + 5.6\ \text{k}\Omega = 9\ \text{k}\Omega$

   $$I_m = \frac{E}{R_T} = \frac{22.5\ \text{V}}{9\ \text{k}\Omega} = \mathbf{2.5\ mA},\ V_m = 2.5\ \text{mA}(2.4\ \text{k}\Omega + 5.6\ \text{k}\Omega) = \mathbf{20\ V}$$

   c. $$V_{3.3\text{k}\Omega} = \frac{3.3\ \text{k}\Omega(12\ \text{V})}{4.5\ \text{k}\Omega} = 8.8\ \text{V}$$

   $V_m = 12\ \text{V} - 8.8\ \text{V} = \mathbf{3.2\ V}$

   $$I_m = \frac{12\ \text{V}}{4.5\ \text{k}\Omega} = \mathbf{2.67\ mA}$$

16. a. $R_T = NR_1 = 8\left(28\frac{1}{8}\ \Omega\right) = 225\ \Omega$

$$I = \frac{E}{R_T} = \frac{120\ \text{V}}{225\ \Omega} = \mathbf{0.53\ A}$$

b. $P = I^2R = \left(\frac{8}{15}\ \text{A}\right)^2\left(28\frac{1}{8}\ \Omega\right) = \left(\frac{64}{\cancel{225}}\right)\left(\frac{\cancel{225}}{8}\right) = \mathbf{8\ W}$

c. $V = IR = \left(\frac{\cancel{8}}{15}\ \text{A}\right)\left(\frac{225}{\cancel{8}}\ \Omega\right) = \mathbf{15\ V}$

d. All go out!

17. $P_s = P_{R_1} + P_{R_2} + P_{R_3}$

$$E{\cdot}I = I^2R_1 + I^2R_2 + 24$$

$$(R_1 + R_2)I^2 - E{\cdot}I + 24 = 0$$

$$6I^2 - 24\,I + 24 = 0$$

$$I^2 - 4\,I + 4 = 0$$

$$I = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(4)}}{2(1)} = \frac{4 \pm \sqrt{16-16}}{2} = \frac{4}{2} = 2\ \text{A}$$

$$P = 24\ \text{W} = (2\ \text{A})^2R, \quad R = \frac{24\ \Omega}{4} = \mathbf{6\ \Omega}$$

18

b. $V_{ab} + 4\ \text{V} + 8\ \text{V} - 16\ \text{V} = 0$, $V_{ab} = 16\ \text{V} - 12\ \text{V} = \mathbf{4\ V}$

c. $V_{ab} + 12\ \text{V} - 18\ \text{V} + 6\ \text{V} - 12\ \text{V} = 0$, $V_{ab} = 30\ \text{V} - 18\ \text{V} = \mathbf{12\ V}$

20. a. $P = 8\text{ mW} = I^2R,\ R = \dfrac{8\text{ mW}}{I^2} = \dfrac{8\text{ mW}}{(2\text{ mA})^2} = \mathbf{2\ k\Omega}$

$$I = \frac{E}{R_T} = \frac{20\text{ V} - E}{3\text{ k}\Omega + 2\text{ k}\Omega} = 2\text{ mA (CW)},\quad E = \mathbf{10\ V}$$

b. $I = \dfrac{16\text{ V}}{2\text{ k}\Omega} = 8\text{ mA},\ R = \dfrac{V}{I} = \dfrac{12\text{ V}}{8\text{ mA}} = \mathbf{1.5\ k\Omega}$

$$I = \frac{E}{R_T} = \frac{E - 4\text{ V} - 10\text{ V}}{2\text{ k}\Omega + 1.5\text{ k}\Omega} = \frac{E - 14\text{ V}}{3.5\text{ k}\Omega} = 8\text{ mA (CCW)}$$

$E = \mathbf{42\ V}$

21

c. $-6\text{ V} - 22\text{ V} - V_1 + 36\text{ V} = 0$

$V = 36\text{ V} - 28\text{ V} = \mathbf{8\ V}$

23

b. $+24\text{ V} - 10\text{ V} - V_1 = 0$

$V_1 = \mathbf{14\ V}$

$+10\text{ V} - V_2 + 8\text{ V} = 0$

$V_2 = \mathbf{18\ V}$

24

b. $+10\text{ V} - V_1 + 6\text{ V} - 2\text{ V} - 3\text{ V} = 0,\ V_1 = \mathbf{11\ V}$

$+10\text{ V} - V_2 - 3\text{ V} = 0,\ V_2 = \mathbf{7\ V}$

27

b. $V = \dfrac{(2\text{ k}\Omega + 3\text{ k}\Omega)(40\text{ V})}{4\text{ k}\Omega + 1\text{ k}\Omega + 2\text{ k}\Omega + 3\text{ k}\Omega} = \dfrac{(5\text{ k}\Omega)(40\text{ V})}{10\text{ k}\Omega} = \mathbf{20\ V}$

c. $\dfrac{(1.5\ \Omega + 0.6\ \Omega + 0.9\ \Omega)(0.72\text{ V})}{(2.5\ \Omega + 1.5\ \Omega + 0.6\ \Omega + 0.9\ \Omega + 0.5\ \Omega)} = \dfrac{(3\ \Omega)(0.72\text{ V})}{6\text{ k}\Omega} = \mathbf{0.36\ V}$

28

c. $$\frac{1000\text{ V}}{100\ \Omega} = \frac{V_2}{68\ \Omega},\ V_2 = \frac{68\ \Omega(1000\text{ V})}{100\ \Omega} = \mathbf{680\ V}$$

$$\frac{1000\text{ V}}{100\ \Omega} = \frac{V_1}{2\ \Omega},\ V_1 = \frac{2\ \Omega(1000\text{ V})}{100\ \Omega} = \mathbf{20\ V}$$

$$E = V_1 + V_2 + 1000\text{ V}$$
$$= 20\text{ V} + 680\text{ V} + 1000\text{ V}$$
$$= \mathbf{1700\ V}$$

d. $V_1 = \mathbf{0\ V}$

$$V_2 = \frac{10\text{ k}\Omega(50\text{ V} - 30\text{ V})}{10\text{ k}\Omega + 3.3\text{ k}\Omega + 4.7\text{ k}\Omega}$$

$$= \frac{10\text{ k}\Omega(20\text{ V})}{18\text{ k}\Omega} = \mathbf{11.11\ V}$$

$$V_x = E_1 - V_{3.3\text{k}\Omega}$$

$$V_{3.3\text{k}\Omega} = \frac{3.3\text{ k}\Omega(20\text{ V})}{18\text{ k}\Omega}$$
$$= 3.67\text{ V}$$

$$V_x = 50\text{ V} - 3.67\text{ V} = \mathbf{46.33\ V}$$

30.

b. $$100\text{ V} = \frac{(6\text{ M}\Omega + R)(140\text{ V})}{6\text{ M}\Omega + R + 3\text{ M}\Omega}$$

$$100(9\text{ M}\Omega + R) = 840\text{ M}\Omega + 140R$$
$$900\text{ M}\Omega + 100R = 840\text{ M}\Omega + 140R$$
$$40R = 60\text{ M}\Omega$$

$$R = \frac{60}{40}\text{ M}\Omega = \mathbf{1.5\ M\Omega}$$

40. a. $V_a = -8\text{ V} + 14\text{ V} = \mathbf{+6\ V}$, $V_b = \mathbf{14\ V}$

$\mathbf{V}_c = +I(10\ \Omega) - 6\text{ V}$ with

$$I = \frac{14\text{ V} + 6\text{ V}}{10\ \Omega + 10\ \Omega} = \frac{20\text{ V}}{20\ \Omega} = 1\text{ A}$$

Therefore, $V_c = (1\text{ A})(10\ \Omega) - 6\text{ V} = 10\text{ V} - 6\text{ V} = \mathbf{4\ V}$

$V_d = \mathbf{0\ V}$

b. $V_{ab} = V_a - V_b = 6\text{ V} - 14\text{ V} = \mathbf{-8\ V}$

$V_{cb} = V_c - V_b = 4\text{ V} - 14\text{ V} = \mathbf{-10\ V}$

$V_{cd} = V_c - V_d = 4\text{ V} - 0\text{ V} = \mathbf{4\ V}$

c. $V_{ad} = V_a - V_d = 6\text{ V} - 0\text{ V} = \mathbf{6\ V}$

$V_{ca} = V_c - V_a = 4\text{ V} - 6\text{ V} = \mathbf{-2\ V}$